



MECH MAINT- TURBINE Department
SIMHADRI

Ref:SMTTPP/MECH MAINT-/2022-23/MM/TMD/130235

Date:

25/05/2022(18:52:02)

Subject:Revision of LMI “Guidance for the removal of seized studs by the drilling/bore welding system” LMI NO. LMI/OGN/OPS/MECH/021- Reg

The LMI: “**Guidance for the removal of seized studs by the drilling/bore welding system with LMI NO. LMI/OGN/OPS/MECH/021**” is revised by CC-OS.

Following changes were done in LMI as per revised OGN:

- Scope of the LMI written in detail
- HOD (MM) changed to HOD (TMD)

Put up for Approval of Competent authority.

Encl:

- Earlier approved LMI.
- New LMI
- Revised OGN

Submitted by:

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The newly revised LMI is Rev-04 May'22 while the earlier one approved in the month of May'22 was Rev-03 May'22 version. There is not much difference between the two except that which has been mentioned in the proposed notesheet (change of **Scope** language in page 2 of 7, and change of Responsibility, **clause no.9**, on page 4 of 7 in Rev-03, May '22). This was necessitated because of the Revision of the OGN by COS - ST, which was received by TMD, post approval of Rev-03 May'22 version.

CA (HOP, Simhadri) may therefore please accord approval for the same i.e for LMI no. LMI/OGN/OPS/MECH/021, **Rev-04 May2022**.

Submitted by:

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कृपया जांच करें।

Please examine.

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Submitted for approval of competent authority

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On:29/05/2022(12:52:58) **Ref:**SMTTPP/MECH MAINT-/2022-23/MM/TMD/130235

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/021/

ISSUE NO.1; REVISION NO. 04; DATE: MAY'2022

**GUIDANCE FOR THE REMOVAL OF
SEIZED STUDS BY THE
DRILLING/BORE WELDING SYSTEM**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	GUIDANCE FOR REMOVAL OF SEIZED STUDS BY DRILLING/BORE WELDING SYSTEM
<u>2.a) DOCUMENT CODE</u> b) REVISION/ISSUE NO.&DATE	<u>LMI/OGN/OPS/MECH/021/</u> <u>ISSUE-1; REV.-04; MAY'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/021, REV:02</u>
<u>4.PRERPARED BY</u>	DGM (TMD)
<u>5. CHECKED BY</u>	HOD (TMD)
<u>6.RECHECKED BY</u>	GM (MAINT)
<u>7. REVIEWED BY</u>	GM (O&M)
<u>8. APPROVED BY</u>	GGM
<u>9.INDEX LIST</u>	ENCLOSED.
<u>10.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: GUIDANCE FOR REMOVAL OF SEIZED STUDS BY DRILLING/BORING SYSTEM		
REV: 04 DATE: MAY'2022		
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<u>2.0 INTRODUCTION</u> During the course of maintenance work in power plant one often comes across the problem of seized studs particularly in those equipment which are exposed to high temperatures. Removal of seized studs especially in turbine cylinders and steam chests is a challenging task and traditionally seized studs are removed either by using considerable force i.e with a large impact/ torque wrench or by drilling out the core of studs and manually picking out the threaded portion of fastener remaining. Old studs are often weakened in-service by the effects of creep and usage of high torque methods may result in fracture of stud. There is also a probability of seized studs picking up and tearing the thread surfaces in the component which would consequently result in additional rectification and longer down time. Through drilling out the core and manually picking out threads is a safe method it requires high level of accuracy as otherwise run out during drilling may result in hard spots or component distortion and cuts into threaded portion. The objective of this LMI is to establish a procedure for removal of seized studs for any item in the plant. <u>3.0 SCOPE</u> The scope of this LMI is to detail the procedure to be adopted for removal of seized steel studs. This LMI applies to steel or Nimonic studs that have seized in their sockets in any Steam Turbine of coal-based power plant. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note OGN/OPS/MECH/021, Rev:02 circulated by operation services-NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> LMI/OGN/OPS/MECH/021 REV-03, DATE: MAY'2022		

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<u>6.0 SPECIFIC INSTRUCTIONS</u> a) Following the procedure detailed below for seized stud removal. b) Exception recording and taking up with manufacturer. <u>7.0 METHODOLOGY OF IMPLEMENTATION, ADVANTAGES AND RESPONSIBILITY CENTERS</u> The maintenance staff is to carry out the following steps: a) Identify and mark the stud which has seized and could not be removed with normal torques. b) Cut off the stud ½” above the joint face using portable saw or disc grinder. c) Using portable radial arm drilling machine or for less accessible areas pneumatic angle drill machine drill down the center of the remaining stud using bolt heating hole as a pilot. d) The drilling to be carried out progressively to 3/16” of thread root using 3 or 4 flute high speed steel core drills. e) Set up and deposit a spiral run of the weld in the bore of the stud. Larger size of studs require slightly heavier deposit of weld. Avoid welding of stud to the component at the bottom of the tapped hole. This welding causes diametrical shrinkage of studs which enables studs to be removed even when the threads have already picked up during initial attempts to screw them out. f) Weld a nut to the top of stud and finally screw them out with a small impact wrench. Advantages: a) Risk of damage to threads of component is reduced with no consequential problems of fitting oversized studs or thread inserts into components, and unexpected costs or programme overruns are avoided. b) Time savings are achieved by using few drilling operations for removing bulk material and in using an automatic welding process. c) Programmes and plans can be maintained by the establishment of consistently achievable bolt replacement timetables. d) Automatic welding machines are flexible in use. e) Specialist services can be avoided since the necessary skills are readily available in power station maintenance teams. Responsibility : HOD (TMD)		

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<u>8. EXCEPTION REPORTING</u> The exceptions encountered during removal of seized studs if any even after following above procedure are to be recorded and taken up with manufacturer. <table border="1"> <thead> <tr> <th>Description of exception</th> <th>Unit No.</th> <th>Location of exception</th> <th>Remarks</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table> <u>9.0 RESPONSIBILITIES</u> The responsibility of carrying out different tasks in relation to removal of seized studs as detailed in the LMI are with HOD of TMD. <u>10. RECORD KEEPING</u> The records to be maintained under this LMI include - compliance record of the procedure detailed under this LMI - Exception reporting The above records are to be maintained by HOD (TMD). <u>11.REVIEW</u> This document shall be reviewed by HOP, SIMHADRI on two-yearly basis.			Description of exception	Unit No.	Location of exception	Remarks				
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NTPC		LMI		SIMHADRI	
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REVISION INDEX (for changes/revision)					
Sl. No.	Rev. no.	Date	Details	Approved by	
1	01	FEB'2013	Point No-7,9,11: Responsibility changed Point No-8: If any added Distribution list: changed		
2	02	JAN'2020	Point no-11: Review period changed to two yearly basis from three-yearly basis. Point no-5: Revision no and date changed Distribution list: changed Compliance record: changed		
3	03	MAY'2020	Point no-7: Advantages of the process as per OGN added. In the advantages, the following point removed as per revised OGN: "Cost savings are achieved by avoiding the need for precision drilling, boring and thread picking operations and the cost of the equipment can often be recovered from one bolting replacement exercise." Responsibility: In point no-7 & 9 all AGMs of MMD is changed to all HODs of MMD.		
4	04	MAY'2022	Scope changed. HOD (MM) changed to HOD (TMD)		

[illegible]

NTPC	COMPLIANCE RECORD		SIMHADRI
TITLE: GUIDANCE FOR REMOVAL OF SEIZED STUDS BY DRILLING/BORING SYSTEM			
REV: 04		DATE: MAY'2022	
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Description of exception	Unit No.	Location of exception	Remarks